

Design and Implementation of High-Performance Multi-Level Approximate Multiplier

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Abstract—In computers and digital electronics, digital multiplication is a fundamental mathematical operation. Because multiplier applications are inherently error-tolerant, approximate multipliers are created. Approximate multipliers are primarily designed to accomplish a compromise between precision and enhanced performance factors like speed, power efficiency, and efficient use of chip area in digital circuit designs. They aim to deliver results that closely approximate the exact multiplication outcome while bringing significant benefits to a wide range of applications. In this study, we present a novel 4-2 approximation compressor architecture that makes use of multiplexers. This is achieved by replacing portions of the normal AND/OR gates with approximate MUX-based logic. The various high-speed 5-3, 10-4, 15-4, and 20-5 compressors are being developed using MUX-based logic. Utilizing the 5-3, 10-4, 15-4, and 20-5 compressor structures, the multiplier's partial products are combined. The circuits under investigation are used to create 8×8 multipliers, 16×16 multipliers, 32×32 multipliers, and 64×64 multipliers implemented in CMOS 45nm technology. The suggested compressor achieves an 11.9% reduction in delay when compared to the precise compressor and the suggested multiplier achieves a 39% reduction in delay. Consequently, this design is well-suited for applications demanding increased processing speed.

Keywords—compressor, Approximate Multiplier, Image processing, Approximate Computing, Digital Arithmetic

I. INTRODUCTION

A basic mathematical process in digital computing is multiplication, it is frequently encountered in tasks ranging from basic arithmetic calculations to complex algorithmic operations. However, traditional multiplication processes can be time-consuming and resource-intensive, particularly when dealing with large binary numbers. This is where a multiplier using a compressor comes into play [1-5]. A multiplier using a compressor is an essential component in digital circuit design, with significant applications in various computing and signal-processing systems [6]. This innovative circuit design is pivotal in enhancing the effectiveness and speed of multiplication operations, making it a fundamental part of modern digital electronics.

The design of approximation multipliers has become an established area of research in recent years [7]. It aligns with the new development in digital design known as approximate computing where the emphasis shifts from demanding precise calculations to prioritizing enhanced speed and power efficiency. So, in contrast to traditional multipliers that prioritize exactness, approximate multipliers intentionally introduce approximations to the multiplication process [4]. These approximations can manifest as simplified logic or reduced bit precision.

Approximate multipliers leverage various techniques, including simplified logic gates, reduced bit precision, and error-tolerant algorithms to expedite the multiplication process while compromising on the exactness of the result. By doing so, they can substantially reduce the hardware complexity, latency, and energy consumption associated with traditional multiplication operations. A compressor, in digital design, is a circuit that takes multiple inputs and produces a smaller set of outputs while preserving certain properties of the inputs [6]-[8]. In an approximate context, an approximate compressor is engineered to provide an output that is a close approximation of the exact result, but with a controlled degree of error. This allows for faster and more power-efficient operation.

The key objective of an approximate multiplier using approximate compressors is to strike a balance between computation accuracy and resource efficiency [9]. By judiciously applying approximate compressors in the multiplication circuitry, it becomes possible to achieve significant reductions in hardware complexity, latency, and power consumption, all while still producing results that fall within an acceptable error margin for the specific application. This approach finds applications in various domains, including digital signal processing, image and video processing, and artificial intelligence, where rapid computation is crucial, and minor deviations from precise results can be tolerated.

II. LITERATURE SURVEY

A. Compressor

In a binary multiplication operation, two binary numbers are multiplied bit by bit, resulting in multiple partial products. Each partial product corresponds to a pair of bits and is generated by multiplying one bit from the first number with all the bits of the second number [2]. Compressors are employed to efficiently reduce and consolidate these partial products [10]. Compressors facilitate the efficient propagation of carry bits, which are generated during the addition of partial products. Efficient carry handling is crucial for preventing propagation delays and ensuring accurate results in high-speed multipliers [11]. Multipliers with compressors are known for their area and power efficiency. By reducing the number of partial products and minimizing the logic gates required for computation, compressors help create compact and energy-efficient multiplier circuits, which is especially important in embedded systems and low-power devices.

B. Exact 4-2 Compressor

The 4-2 compressor is a digital circuit that takes 4 input bits and produces two output bits in compressed form. The Exact compressor is shown in Fig.1. This design uses full adders connected in a cascaded manner with less carry propagation. Here x_1 , x_2 , x_3 , x_4 and C_{in} are taken as the

inputs of the Exact 4-2 compressor, $Cout$, $Carry$ and Sum are taken as the outputs [12]. $Cout$, $Carry$ and Sum are provided accordingly

$$Sum = x1 \oplus x2 \oplus x3 \oplus x4 \oplus Cin \quad (1)$$

$$Cout = (x1 \oplus x2) \cdot x3 + (x1 \oplus x2)' \cdot x1 \quad (2)$$

$$Carry = (x1 \oplus x2 \oplus x3 \oplus x4) \cdot cin + (x1 \oplus x2 \oplus x3 \oplus x4) \cdot x4 \quad (3)$$

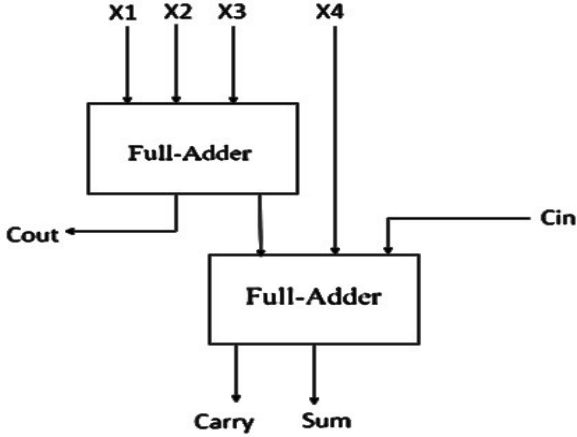


Fig. 1. Exact4:2 compressor.

C. Approximate 4-2 Compressor

An Approximate 4-2 compressor sacrifices a degree of accuracy in its compression operation for various practical benefits such as reduced hardware complexity and improved speed, or power efficiency, which makes it suitable for specific applications where strict precision isn't a primary concern. The Approximate 4-2 compressor is shown in Fig.2. In this context, the approximation process involves the elimination of the Carry-out ($Cout$) signal [13]-[15]. However, it's important to note that an error has taken place. specifically, if the input combination is '1111.' In this scenario, where all inputs are set to high, the $Carry$ and Sum outputs are configured as '11,' and this introduces an error with a value of -1. $Carry$ and Sum are given as

$$Sum = (x4 \cdot x3) + (x4 \cdot x1)' \cdot x2' + (x1' \cdot x2 \cdot x3) + (x1 \cdot x2 \cdot x4) + (x1 \cdot x2 \cdot x3) + (x1 \cdot x2' \cdot x3' \cdot x4') + (x1' \cdot x2) + (x3' \cdot x4') \quad (4)$$

$$Carry = (x1 \cdot x2)' \cdot (x3 \cdot x4)' \quad (5)$$

D. Proposed 4-2 Compressor Using Mux

An approximate 4-2 compressor design using multiplexers (MUX) can be achieved by replacing some of the regular AND/OR gates with approximate MUX-based logic. Here's a high-level description of the design: Step 1: Replace the AND/OR gates with approximate MUX-based logic. Replace some of the regular AND/OR gates in the G and P signal generation with approximate MUX-based logic [12].

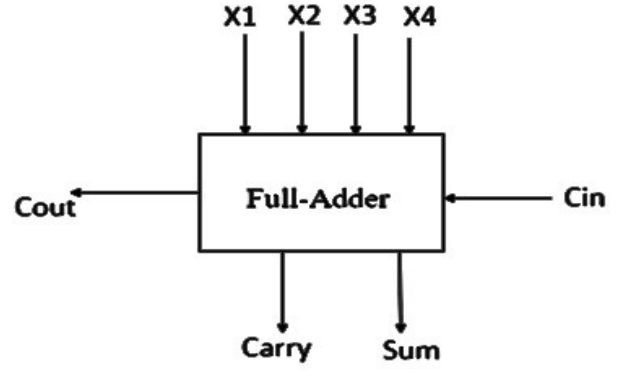


Fig. 2. Approximate 4:2 compressor

For example, you can use MUX'es with nontraditional input selections or unequal input sizes, which introduce controlled levels of inaccuracy in the logic operation. Step 2: Connect the outputs of the Multiplexer-based logic to generate the sum bits. Connect the outputs of the approximate Multiplexer-based logic to generate the approximate sum bits. The specific implementation details of the approximate Multiplexer-based logic would depend on the approximation techniques used and the desired trade-off between accuracy and hardware requirements. It's important to carefully evaluate the accuracy and performance characteristics of the approximate Multiplexer-based logic to ensure that the overall system meets the desired specifications for the intended application. Additionally, appropriate testing and verification should be performed to validate the functionality and performance of the approximate 4-2 compressor design.[15]-[18] So, any type of logic gate can be realized by using a Multiplexer. This will make an impact on the area and delay of existing compressor design. By using the Multiplexer 4-2 compressor designed below Fig.3. Compared to alternative compressor adder architectures now in use, the compressor adder design speeds up performance and decreases the crucial latency in multiplier units.

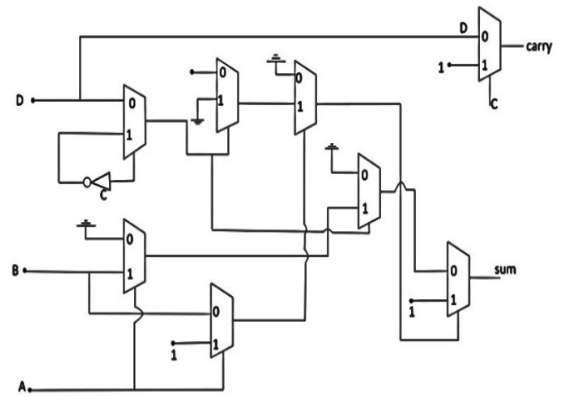


Fig. 3. Proposed Approximate 4:2 compressor

E. Implementation of Approximate 5:3 Compressor

The hardware layout of the proposed 5-3 compressor adder is utilized in the multiplier. Utilizing logic gates, half-adders, and full adders, a new 5-3 compressor adder is created. The adder circuit accepts five inputs [PR0-PR4] and adds them using two mux-based half-adders and two mux-based full-adders, producing an output of three bits [S0-S3] as shown in Fig.4.

F. Implementation of Approximate 10:4 Compressor

Fig. 5 illustrates the 16-bit Proposed multiplier that utilizes a 10-4 compressor system architecture. A proposed 10-4 compressor takes ten inputs (PR0–PR9), adds them with two proposed 5-3 compressors, and outputs four bits (S0–S3). Two sets of proposed 5-3 compressors, as well as two sets of mux-based full and mux-based half adders [7], were used in the construction of this compressor circuit. It can produce a maximum output of 1010. The 5-3 compressor speeds up the multiplier while reducing latency and gate counts in a 10-4 compressor circuit. One 5-3 compressor and another 5-3 compressor circuit are used in the 10-4 compressor to add the first five inputs [PR0 - PR4] and the next five inputs [PR5 - PR9]. From the 5-3 compressors, the final 4-bit output [S0 - S3] is obtained.

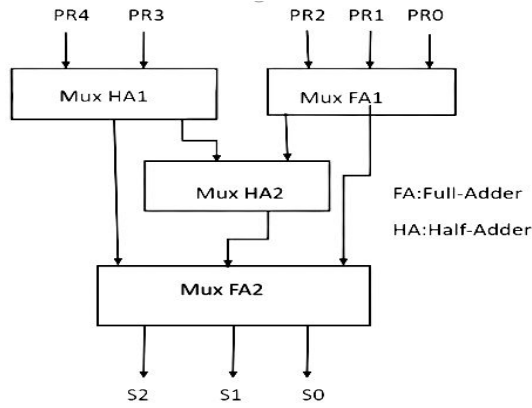


Fig. 4. Proposed Approximate 5:3 compressor

G. Implementation of Proposed Approximate 15-4 compressor

An adder circuit known as a 15-4 compressor [7] adds 15 bits [PR0–PR14] at a time, producing four consequent bits [S0–S3] as shown in Fig.6. A parallel adder, two proposed 5-3 compressors and five mux-based full adders are used to design the 15-4 compressor circuit [10]. The 15-4 compressor maximum output will be 1111 as a result [7]. To reach the final result [S0 - S3], a 4-bit parallel adder is also used to combine the outputs of two 5-3 compressors.

H. Implementation of Proposed Approximate 20-5 compressor

A 20-5 compressor circuit produces the five subsequent bits (S0-S4) shown in Fig.7 by adding 20 bits (PR0-PR19) at a time.[10]-[16] The compressor circuit consists of two MUX-based full adders, two mux-based half adders, and one mux-based full adder, as well as one proposed 5-3 compressor and one proposed 15-4 compressor.

III. PROPOSED APPROXIMATE MULTIPLIER

Multipliers require more time and more energy. The three phases of competition are creating a partial product, reducing the Partial product, and adding it to the final phase. It will consume more power and energy to reduce the product. Here, multipliers are implemented with $n = 8$ and $n = 16$, $n = 32$, $n = 64$, and four configurations are considered for each multiplier using the approximate 4-2, 5-3, 10-4, 15-4, 20-5 compressors. To minimize the number of partial products, a compressor is employed [20]. A compressor, as mentioned earlier, takes multiple binary inputs and produces a compressed version with fewer bits while trying to preserve the most significant information. This results in the maximum

heights for the various stages being 2, 4, 8, 16, and 32, due to the usage of compressors. Only when it is necessary to reduce the PPM to the desired level do full and half adders come into play.[17]-[20] Fig. 8 shows the reduction strategy for a setup of unsigned 8×8 multipliers as an example. similarly done for all the other sizes. In Fig.8 you can see that the connection between the carry and sum outputs of the approximately 4-2 compressor is represented by the dashed line, while the solid lines used to form the connection between the outputs of the approximately 4-2 compressor to the Full adder [20] The connections between the Tin-Tout pins of approximately 4 2 compressors are indicated by arrows.

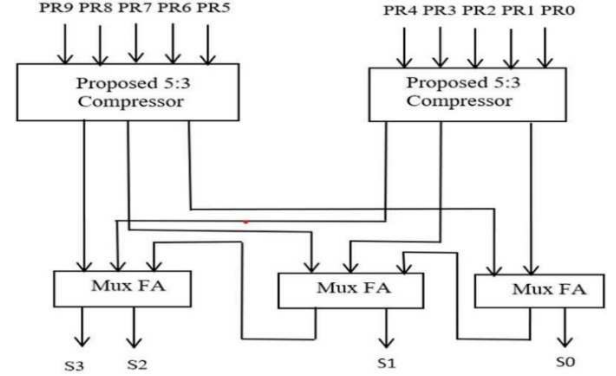


Fig. 5. Proposed Approximate 10:4 compressor

IV. RESULTS AND ANALYSIS

A. Simulation of compressors

In this study, we present the schematics and simulation waveform results of the proposed 4:2 compressors in Fig.9 and Fig .10, respectively same as the results of the proposed 20:5 compressor present in Fig.11 and Fig.12. These visual representations were generated using Xilinx software version 14.7. The waveform illustrates that there is only one combination with an error, resulting in a marginal decrease in accuracy of just 0.5%. To evaluate performance metrics like Power consumption, speed, and area utilization, we employed Cadence Genus. All of our designs were developed using a 45nm technology node, and we obtained the output data using a standardized Cadence library, as detailed in the following TABLE I. Here we can see from TABLE I and TABLE II that the introduced compressor has significantly reduced area, power, and latency compared to the Exact compressor. Here is a comprehensive analysis of various methodologies to assess how they demonstrate improvements in area efficiency, power efficiency, delay reduction, and switching activity compared to a precise 4:2 compressor.

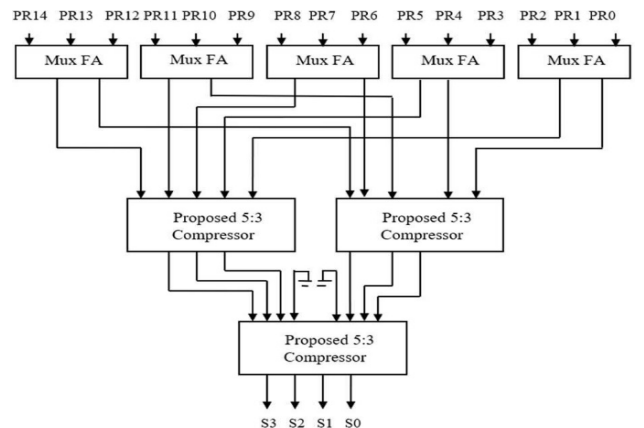


Fig. 6. Proposed Approximate 15:4 compressor

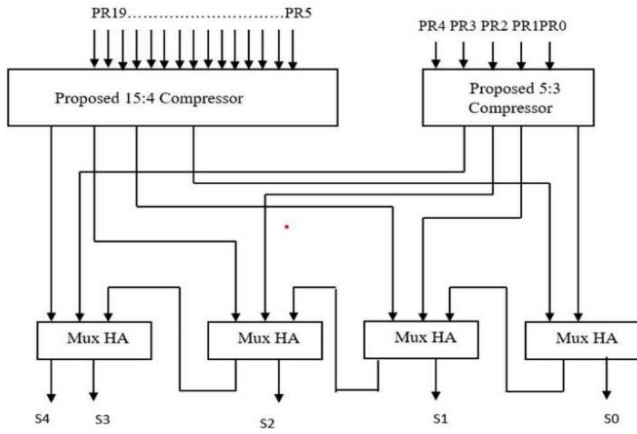


Fig. 7. Proposed Approximate 20:5 compressor

B. Simulation of Approximate Multipliers

The Simulation Results presented in Fig.13 and Fig.14 show the schematic results of the 64×64 multiplier designed using a proposed compressor. The multiplier's delay is minimized through the implementation of the proposed compressors due to the reduction in switching activity. In comparison to the prior multiplier, delay is minimized through the implementation of the proposed compressors due to the reduction in switching activity. In comparison to prior multiplier designs, the proposed multipliers feature a notably lower count of LUTs, causing for decrease in both area and power consumption. Table III shows the metrics of approximate multipliers for different sizes. As expected, multipliers using the accurate 4:2 compressor have a large delay and consume more power than the multipliers using the Approximate compressors.

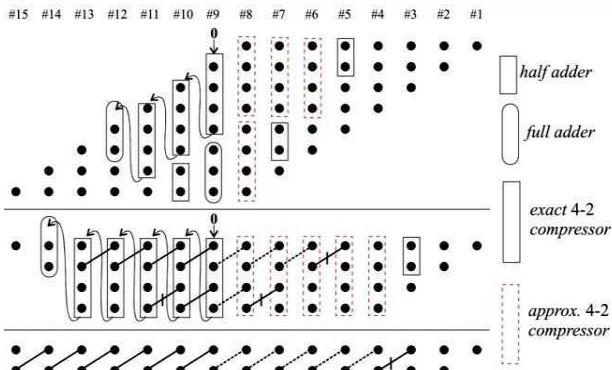


Fig. 8. Proposed Approximate Multiplier

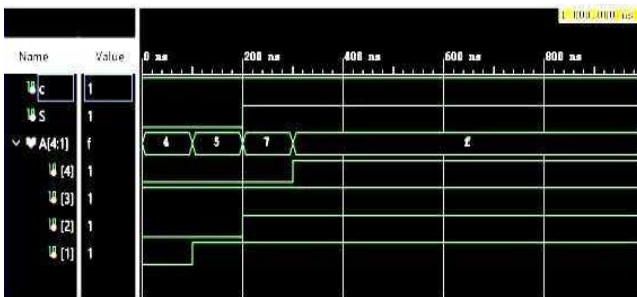


Fig. 9. The Proposed Approximate compressor

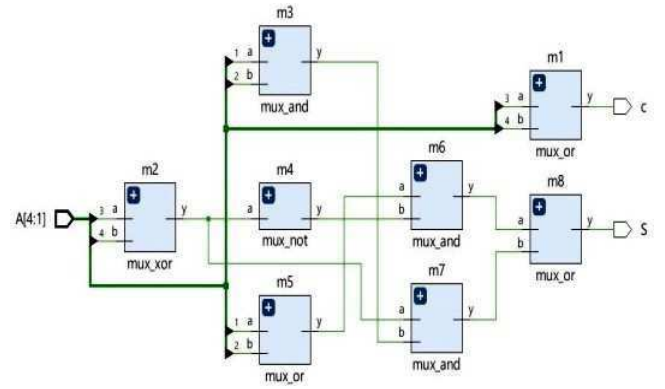


Fig. 10. Schematic of Proposed Approximate Compressor

This newly proposed compressor demonstrated an impressive 11% reduction in delay as opposed to the Exact Compressor, indicative of its heightened performance. Subsequently, the compressor was extended to higher bit configurations, namely 5:3, 10:4, 15:4, and 20:5, catering to enhanced computational efficiency in larger multipliers such as 32×32 and 64×64 . In the ensuing analysis, the proposed compressor was systematically juxtaposed against the exact compressor, evaluating metrics of Area, Power, and Timing. The primary objective of this research is to expedite multiplier computations while accepting a minimal error rate. The implemented compressors (4:2, 5:3, 10:4, 15:4, and 20:5) were incorporated into different-sized multipliers, and the ensuing results were contrasted with exact multipliers with discernible distinctions. For a 64×64 multiplier, the proposed compressors exhibited a substantial reduction in Area (40%), Power (38%), and delay (39%). Future endeavors will focus on further diminishing the error rate to enhance the proposed compressors' performance, approaching that of exact compressors. The genus Tool was employed for the comprehensive evaluation of Area, Power, and Timing parameters.

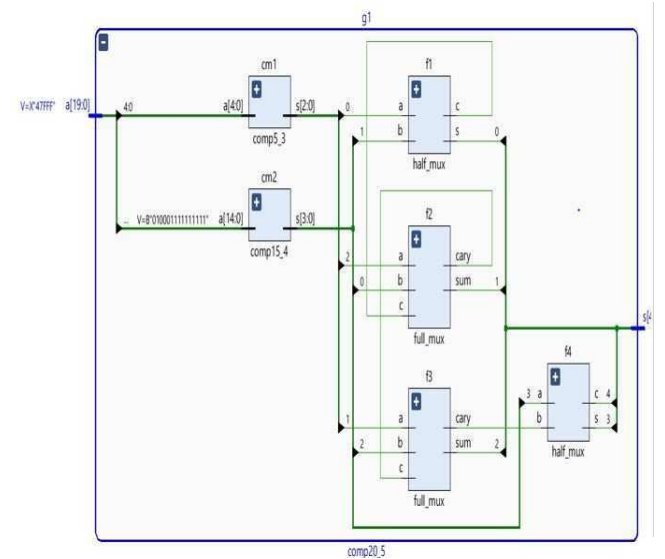


Fig. 11. Schematic of Proposed Approximate 20-5 compressor

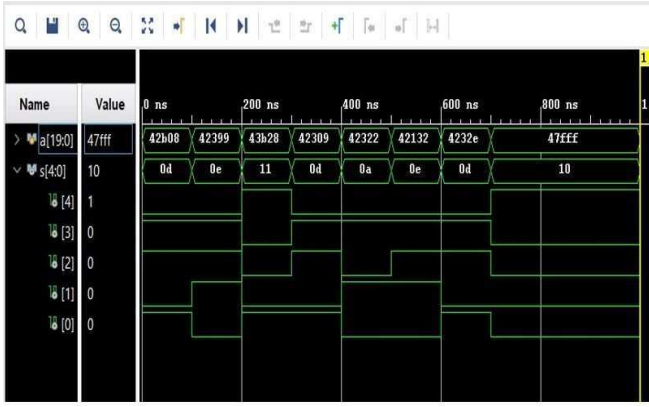


Fig. 12. Simulation result of the proposed Approximate 20-5 compressor

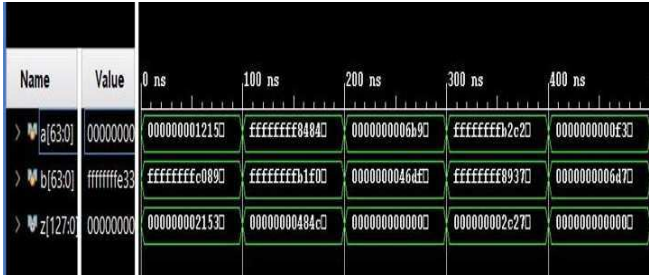


Fig. 13. Simulation result of Proposed 64x64 Approximate Multiplier

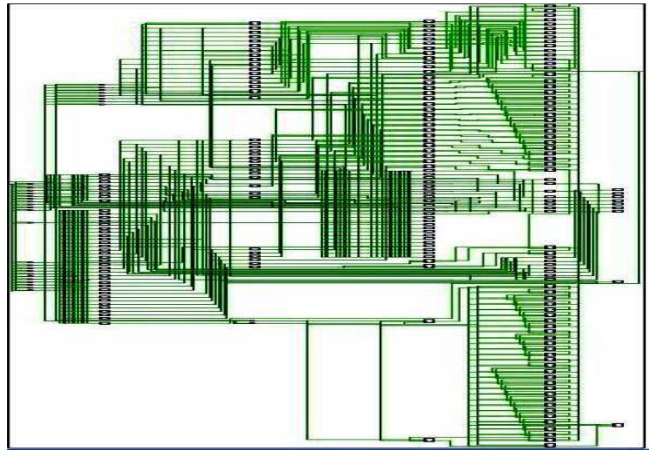


Fig. 14. Schematic of Proposed Approximate 64x64 multiplier

TABLE I. COMPARISON OF PROPOSED APPROXIMATE COMPRESSOR

	Exact compressor	Previously Proposed compressor	Proposed compressor
Area	39.359 (100%)	27.248 (9%)	21.950(11%)
Power	$0.910 \times e^{-06}$	$0.748 \times e^{-06}$ (47%)	$0.6684 \times e^{-06}$ (57%)
Timing(ps)	1659	1459(46%)	1266(53%)
Error Rate	0	3	1

V. CONCLUSION

In this project, the proposed multiplier using the compressor has less power, delay, and area and also has a high performance when compared to the exact compressors. In this project, we have introduced a novel compressor design that offers significant advantages over traditional compressors. Our proposed Approximate 4-2 compressors, along with various compressor sizes, have undergone thorough investigation and have been implemented in a 45nm CMOS technology for use in Approximate Multipliers. Compared to previous multipliers that utilized an approximate tree

compressor, our proposed compressor has achieved a remarkable 39% reduction in delay. While this enhancement has resulted in a slight decrease in accuracy by only 0.5%, it positions our multiplier as an excellent choice for applications requiring swift processing speeds. Furthermore, our innovative design exhibits a remarkable 40% reduction in consumption of power and a 38% decrease in area requirements. These characteristics make the solution particularly suitable for applications like image contrast adjustment and multimedia processing, where power efficiency and compactness are crucial. Approximate multipliers have a promising future scope, especially in fields where high-performance computation is required with a tolerance for errors or inaccuracies.

TABLE II. COMPARISON OF DIFFERENT SIZES PROPOSED APPROXIMATE COMPRESSORS

	Area	Power	Timing (ps)
Exact Compressor 5:3	60.560	$4.276 \times e^{-06}$	1692
Proposed Compressor 5:3	43.900	$1.39 \times e^{-06}$	1591
Exact Compressor 10:4	190.340	$15.27 \times e^{-06}$	2772
Proposed Compressor 10:4	127.916	$13.83 \times e^{-06}$	2027
Exact Compressor 15:4	298.90	$7.107 \times e^{-06}$	2640
Proposed Compressor 15:4	241.451	$8.274 \times e^{-06}$	2297
Exact compressor 20:5	402.505	$14.55 \times e^{-06}$	3134
Proposed Compressor 20:5	322.439	$12.87 \times e^{-06}$	4325

TABLE III. COMPARISON OF PROPOSED APPROXIMATE MULTIPLIER

	Area	Power(w)	Timing (ps)
Accurate Multiplier (4x4)	278.539	15682.3(nw)	3330
Proposed Approximate Multiplier (4x4)	127.96 (45%)	127.916(nw) (48%)	1970 (37%)
Accurate Multiplier (8x8)	1837.330	$5.9521 \times e^{-06}$	4100
Proposed Approximate Multiplier (8x8)	1127.024 (37%)	$4.5359 \times e^{-06}$ (36%)	3730(39%)
Accurate Multiplier (16x16)	5490.00	$39.16 \times e^{-06}$	8985
Proposed Approximate Multiplier (16x16)	4218.0 (38%)	$14.41 \times e^{-06}$ (48%)	8226(47%)
Accurate Multiplier (32x32)	1096.625	$74.440 \times e^{-06}$	4517
Proposed Approximate Multiplier (32x32)	1089.791(40%)	$70.71 \times e^{-06}$ (40%)	4096 (40%)
Accurate Multiplier (64x64)	1837.330	$1156.4 \times e^{-06}$	8990
Proposed Approximate Multiplier (64x64)	1034.024 (38%)	$226.5 \times e^{-06}$ (40%)	5385 (37%)

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